

MUHAMMAD FARHAN

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Backend Developer with 4 years of experience designing and building scalable web applications, high-performance APIs, and distributed systems. Proficient in Node.js, TypeScript, and Go, with hands-on experience using frameworks such as Express.js, Go-Chi, and Gin. Experienced in system modernization, microservices architecture, performance optimization, and secure backend development. While my primary focus is backend engineering, I also have practical experience with React.js, enabling effective collaboration across full-stack teams and contributing to the delivery of reliable, scalable, and business-oriented software solutions.

EXPERIENCE

BACKEND DEVELOPER

PT Inovasi Solusi Internasional | Jakarta, Indonesia

2022 – PRESENT

- a. Regol (Forex Client Onboarding System)
 - Spearheaded the complete backend migration of a high-traffic Forex client registration platform from legacy PHP to TypeScript.
 - Identified and resolved critical legacy bugs, ensuring 100% functional parity with the previous system while enhancing codebase reliability.
 - Optimized system performance, successfully reducing overall server resource utilization by up to 75% during off-peak hours.
- b. Eform (Internal CRM & Customer Management)
 - Executed the strategic rewrite of an internal customer management system from PHP/Laravel to TypeScript and Express.js.
 - Mitigated numerous security vulnerabilities and bugs identified in penetration testing reports, significantly hardening the application's security posture.
 - Implemented a dynamic, granular Role-Based Access Control (RBAC) system allowing clients to configure permissions via a dashboard, which included automated data masking/redaction on sensitive documents for unauthorized roles.
 - Engineered the migration with zero downtime, introduced new client-requested features, and achieved a 30% reduction in computing resource usage.
- c. Dynamic Object Detection System
 - Developed a dynamic Computer Vision system leveraging Deep Learning with YOLOv8 for real-time object detection and automated bounding box rendering.
 - Designed a flexible video ingestion pipeline supporting multiple, dynamic input sources switched on-the-fly by users, including remote CCTV streams, direct server webcams, and client-side webcams.
- d. Etrade (Forex Trading & Microservices Architecture)
 - Collaborated on rebuilding a core Forex trading system from scratch using Express.js and JavaScript under strict constraints with zero access to the legacy vendor's locked source code or database.
 - Designed and integrated multiple independent microservices to decouple complex trading functionalities and improve system resilience.
 - Revamped data aggregation pipelines used for analytics, successfully dropping query response times from over 30 seconds to under 3 second.
 - Built entirely new system modules and successfully modernized a legacy desktop-based exchange reporting application into a scalable web platform.

- e. Blockchain Payment Gateway
 - Engineered a dynamic multi-chain crypto payment gateway supporting USDT across Ethereum, Binance Smart Chain (BSC), Tron, and other networks.
 - Developed automated on-chain verification mechanisms that monitor transaction confirmations and block depths to prevent double-spending and ensure payment finality.
 - Created an abstract configuration layer allowing dynamic fee management and seamless addition of new blockchains or tokens, delivering a frictionless user experience akin to QRIS scanning.

EDUCATION

Binus University (Online Learning) – Computer Science

AUGUST 2023 - PRESENT

- Current Status: 6th Semester | GPA: 4.0 / 4.0
- Specialization (Streaming): Artificial Intelligence
- Core Computer Science coursework: Algorithm & Programming Foundations, Operating Systems (Threads & CPU Cycles), Discrete Mathematics, Calculus, and Dynamic Programming.
- AI & Data Science coursework: Natural Language Processing (NLP), Machine Learning, Computer Vision Applications, and Data Science.

SKILLS

- Programming Languages: JavaScript, TypeScript, Go (Golang), SQL, HTML/CSS.
- Backend Development & Frameworks: Node.js, Express.js, Go-Chi, Gin Gonic, Microservices Architecture, RESTful APIs, Role-Based Access Control (RBAC).
- Database & Tools: PostgreSQL, MySQL, Redis, ElasticSearch.
- Developer Tools: Git, Linux/Unix Environments, vibe coding tools (Claude Code and Antigravity CLI).
- Good communication
- Core Competencies: System Modernization & Migration, Database Optimization, Security Hardening & Vulnerability Mitigation, Reverse Engineering, Agile/Scrum Development, Technical Problem Solving, Cross-functional Collaboration.